Q1 - Domain and Range

The domain of definition of function $f(x)=\sqrt{\sin^{-1}(2x)+\frac{\pi}{6}}$ for real values of x, is

(1) $\left[-\frac{1}{4}, \frac{1}{2}\right]$
(2) $\left[-\frac{1}{2}, \frac{1}{2}\right]$
(3) $\left(-\frac{1}{2}, \frac{1}{4}\right)$
(4) $\left[-\frac{1}{4}, \frac{1}{4}\right]$

Q2 - Domain and Range

Let $f:(-1,1) \rightarrow R$ be such that $f(\cos 4\theta)=\frac{2}{2-\sec^2 \theta}$ for $\theta \in\left(0, \frac{\pi}{4}\right) \cup\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$, then the value(s) of $f\left(\frac{1}{3}\right)$ is (are)

(1) $2+\sqrt{\frac{3}{2}}$
(2) $1+\sqrt{\frac{3}{2}}$
(3) $1-\sqrt{\frac{2}{3}}$
(4) $1+\sqrt{\frac{2}{3}}$

Q3 - Domain and Range

If $x \in R$, then the value of expression $\frac{x^2+14x+9}{x^2+2x+3}$ lies between

(1) 5 and 4

(2) -5 and 4

(3) $(-\infty,-4) \cup(5, \infty)$

(4) None of these

Q4 - Domain and Range

The range of $f(x)=\log_{\sqrt{5}}\{\sqrt{2}(\sin x-\cos x)+3\}$ is

(1) [0,1]

(2) [0,2]

(3) $\left[a\frac{3}{2}\right]$

(4) None of these

Q5 - Identical Functions

Which of the following pairs of functions are identical?

(1) $f(x)=\sin^{-1}x+\cos^{-1}x$ and $g(x)=\frac{\pi}{2}$

(2) $f(x)=\tan^{-1}x+\cot^{-1}x$ and $g(x)=\frac{\pi}{2}$

(3) $f(x)=\sec^{-1}x+\operatorname{cosec}^{-1}x$ and $g(x)=\frac{\pi}{2}$

(4) All of the above

Q6 - Identical Functions

Which of the following pairs of functions are identical?

(1) $f(x)=\sqrt{x^2}$ and $g(x)=(\sqrt{x})^2$

(2) $f(x)=\sec\left(\sec^{-1}x\right)$ and $g(x)=\operatorname{cosec}\left(\operatorname{cosec}^{-1}x\right)$

(3) $f(x)=\sqrt{\frac{1+\cos 2x}{2}}$ and $g(x)=\cos x$

(4) $f(x)=x$ and $g(x)=e^{\log_\theta x}$

Q7 - Identical Functions

Which of the following pairs of functions are identical?

(1) $f(x)=\sqrt{1+\sin x}$ and $g(x)=\sin\frac{x}{2}+\cos\frac{x}{2}$

(2) $f(x)=\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ and $g(x)=2\tan^{-1}x$

(3) $f(x)=\sqrt{x^2}$ and $g(x)=(\sqrt{x})^2$

(4) $f(x)=\log_e x^3+\log_e x^5$ and $g(x)=8\log_e x$

Q8 - Periodic Function

The period of $f(x)=\sin^{-1}(\sin x)$ is

(1) π

(2) 2π

(3) $\frac{\pi}{2}$

(4) None of these

Q9 - Periodic Function

The period of $f(x)=|\cos 2x|$ is

(1) $\frac{\pi}{2}$

(2) π

(3) 2π

(4) None of these

Q10 - Periodic Function

The period of $f(x)=|\sin x|+|\cos x|$ is

(1) $\frac{\pi}{2}$

(2) π

(3) 2π

(4) None of these

Q11 - Periodic Function

The period of $f(x)=\sin^4 x+\cos^4 x$ is

(1) π

(2) $\frac{\pi}{2}$

(3) 2π

(4) None of these

Q12 - Periodic Function

The period of $f(x)=\sec(\sin x)$ is

(1) $\frac{\pi}{2}$

(2) 2π

(3) π

(4) None of these

Q13 - Periodic Function

If $f(x)=\sin\sqrt{[a]}x$, where $[\cdot]$ denotes the greatest integer function, has π as its period, then

(1) $a=1$

(2) $a=9$

(3) $a\in[1,2)$

(4) $a\in[4,5)$

Q14 - Periodic Function

The fundamental period of

$f(x)=x+a-[x+b]+\sin\pi x+\cos 2\pi x+\sin 3\pi x+\cos 4\pi x+\cdots+\sin(2n-1)\pi x+\cos 2n\pi x$ for every $a,b\in R$ (where, $[\cdot]$ denotes greatest integer function), is

(1) 2

(2) 4

(3) 1

(4) 3

Q15 - Periodic Function

The period of $f(x)=\sin\frac{\pi}{4}[x]+\cos\frac{\pi x}{2}+\cos\frac{\pi}{3}[x]$, where, $[\cdot]$ denotes greatest integer function, is

(1) 8

(2) 12

(3) 24

(4) None of these

Q16 - Periodic Function

The period of function $f(x)=[x]+\left[x+\frac{1}{3}\right]+\left[x+\frac{2}{3}\right]-3x+15$ is

(1) $\frac{1}{3}$

(2) $\frac{2}{3}$

(3) 1

(4) non-periodic

Q17 - Composite Functions

If $f:\{-6,6]\rightarrow R$ is defined by $f(x)=x^2-3$ for $x\in R$, then (fofof)(-1)+ (fofof) (0)+ (fofof) (1) is equal to

(1) $f(4\sqrt{2})$

(2) $f(3\sqrt{2})$

(3) $f(2\sqrt{2})$

(4) $f(\sqrt{2})$

Q18 - Composite Functions

If $g(x)=x^2+x-2$ and $\frac{1}{2}(gof)(x)=2x^2-5x+2$, then $f(x)$ is equal to

(1) $2x-3$

(2) $2x+3$

(3) $2x^2+3x+1$

(4) $2x^2-3x-1$

Q19 - Composite Functions

Let f and g be the functions defined by $f(x)=\frac{x}{x+1}, g(x)=\frac{x}{1-x}$, then $(f\circ g)(x)$ is equal to

(1) $\frac{1}{x}$

(2) $\frac{1}{x-1}$

(3) $x-1$

(4) x

Q20 - Composite Functions

Let $g(x)=1+x-[x]$ and $f(x)=\begin{cases}-1, & x<0\\ 0, & x=0,\\ 1, & x>0\end{cases}$ then for all x, $f\{g(x)\}$ is equal to

(1) x

(2) 1

(3) $f(x)$

(4) $g(x)$

Q21 - Composite Functions

Let $f(x)$ be a function whose domain is [-5,7] and $g(x)=|2x+5|$. Then, domain of $(fog)(x)$ is

(1) [-4,1]

(2) [-5,1]

(3) [-6,1]

(4) None of these

Q22 - Composite Functions

If $f(x)=\cot^{-1}x : R^{+}\rightarrow\left(0,\frac{\pi}{2}\right)$ and $g(x)=2x-x^{2} : R\rightarrow R$. Then, the range of function $f\{g(x)\}$ wherever define is

(1) $\left(0,\frac{\pi}{2}\right)$

(2) $\left(0,\frac{\pi}{4}\right)$

(3) $\left[\frac{\pi}{4},\frac{\pi}{2}\right)$

(4) $\left\{\frac{\pi}{4}\right\}$

Q23 - Mappings

If for two functions g and f, gof is both injective and surjective, then which of the following is true?

(1) g and f should be injective and surjective

(2) g should be injective and surjective

(3) f should be injective and surjective

(4) None of the above

Q24 - Mappings

Let R and C denotes the set of real numbers and complex numbers respectively. The function $f: C \rightarrow R$ defined by $f(z)=|z|$ is

(1) one-one

(2) onto

(3) bijective

(4) neither one-one nor onto

Q25 - Mappings

Let $f(x)=\frac{x^2-4}{x^2+4}$ for $|x|>2$, then the function $f:(-\infty,-2] \cup[2, \infty) \rightarrow(-1,1)$ is

(1) one-one into

(2) one-one onto

(3) many-one into

(4) many-one onto

Q26 - Mappings

A function f from the set of natural numbers to integers defined by $f(n)=\begin{cases}\frac{n-1}{2}, & \text{if } n \text{ is odd} \\ \frac{n}{2}, & \text{if } n \text{ is even}\end{cases}$, is

(1) one-one but not onto

(2) onto but not one-one

(3) one-one and onto

(4) neither one-one nor onto

Q27 - Mappings

Let $f(x)=x+\sqrt{x^2}$ is a function from $R \rightarrow R$, then $f(x)$ is

(1) injective

(2) surjective

(3) bijective

(4) None of these

Q28 - Mappings

A mapping from N to N is defined as follows :

$f: N \rightarrow N, f(n)=(n+5)^2, n \in N$. Then, f is

(1) one-one

(2) onto

(3) one-one and onto

(4) one-one but not onto

Q29 - Inverse of Function

Which of the following functions have inverse function?

(1) $f(x)=\frac{1}{x-1}, f: R-\{1\} \rightarrow R-\{0\}$

(2) $f(x)=x^2, f: R \rightarrow R$

(3) $f(x)=x^2, f: R^{+} \rightarrow R$

(4) $f(x)=x^2, f: R^{-} \rightarrow R$

Q30 - Inverse of Function

Let $f:(2,4) \rightarrow(1,3)$ be a function defined by $f(x)=x-\left[\frac{x}{2}\right]$ (where, $[\cdot]$ denotes greatest integer function), then $f^{-1}(x)$ is equal to

(1) $2x$

(2) $x+\left[\frac{x}{2}\right]$

(3) $x+1$

(4) $x-1$

Q31 - Inverse of Function

Let $f:[-\sqrt{2}+1, \sqrt{2}+1] \rightarrow\left[\frac{-\sqrt{2}+1}{2}, \frac{\sqrt{2}+1}{2}\right]$ be a function defined by $f(x)=\frac{1-x}{1+x^2}$, then $f^{-1}(x)$ is

(1) $\sqrt{4x-4x^2+1}, x \neq 0$

(2) $\begin{cases} \dfrac{-1+\sqrt{4x+1-4x^2}}{2x}, & x \neq 0 \\ 1, & x=0 \end{cases}$

(3) does not exist

(4) None of these

Q32 - Inverse of Function

Let $a>1$ be a real number and $f(x)=\log_a x^2$, for $x>0$. If $f^{-1}(x)$ is the inverse function of f and b, c are real numbers, then $f^{-1}(b+c)$ is equal to

(1) $f^{-1}(b) \cdot f^{-1}(c)$

(2) $f^{-1}(b)+f^{-1}(c)$

(3) $\frac{1}{f(b+c)}$

(4) None of these

Answer Key

Q1 (1)	**Q2** (2)	**Q3** (2)	**Q4** (2)
Q5 (2)	**Q6** (2)	**Q7** (4)	**Q8** (2)
Q9 (1)	**Q10** (1)	**Q11** (2)	**Q12** (3)
Q13 (4)	**Q14** (1)	**Q15** (3)	**Q16** (1)
Q17 (1)	**Q18** (1)	**Q19** (4)	**Q20** (2)
Q21 (3)	**Q22** (3)	**Q23** (1)	**Q24** (4)
Q25 (3)	**Q26** (3)	**Q27** (4)	**Q28** (4)
Q29 (1)	**Q30** (3)	**Q31** (2)	**Q32** (1)